CS ELEC 6 – Introduction to Cognitive Computing

Prefinal Project Proposal

COGNITIVE CAMPUS ASSISTANT

An Enrollment Requirements and Student Services Advisor for [School name] Registrar's Office

Team: [Team name]

Member A – Knowledge and conversation: [Full name]

Member B – Data and learning: [Full name]

Member C – Integration and interface: [Full name]

Member D – Perception and analytics: [Full name]

BSCS, School of Engineering and Computer Studies · CS4-A · 1st Semester, AY 2026–2027

Instructor: Neal Errol K. Bulawan

Date submitted: [date]

PART A – THE SITUATION

1. Title Page

The title page precedes this section and lists the system title, Members A to D, the course, the instructor, and the date.

2. The Situation

The place and the people. The Registrar's Office of [School name] serves freshmen, transferees, and returning students every enrollment period. Two to [number] staff members answer questions at the window, by Messenger, and by phone. Most of the answers are already printed in the student handbook and on posted announcements, but students still ask staff in person.

How the work is done today. A student joins the queue, states the concern, and the staff member recalls the requirements for that type of student, checks the documents the student carries, and tells the student which window to visit next. The same steps are repeated for each student.

What goes wrong. The team's observation as clients of the service (to be completed with the numbers collected) points to these problems:

- Repeated questions: the same five or six questions (requirements, schedule, fees, clearance, office hours) make up most of the queue.
- Inconsistent answers: different staff members or older posts give different wording or outdated dates.
- Wasted trips: students arrive with incomplete documents and are sent back.
- Staff overload at peak days, with no record of which questions are most common, so staff cannot plan ahead.

Evidence the team collects (public information and own experience only)	Value
Queue observations (number of visits, date, time of day)	[n visits; dates]
Average waiting time observed	[minutes]
Share of questions that were repeats of the five common topics	[%, from n questions noted]
Number of times a student was sent back for missing documents	[count]
Source of each number	[team log, date, who recorded]

3. Users and Their Needs

Role	User story
Freshman	As a freshman, I can ask what documents I need to enroll so that I do not make a wasted trip.
Transferee	As a transferee, I can tell the assistant what documents I already have and learn exactly which are missing.
Returning student	As a returning student, I can ask about clearance steps and deadlines so that I enroll on time.
Any student (Filipino or English)	As a student, I can type or speak in English, Filipino, or a mix, and still get a correct answer.
Any student	As a student, I can see why the assistant gave an answer and reach a real staff member when it is unsure.
Registrar staff	As a staff member, I can see which questions are most common and which the assistant could not answer, so that I can update announcements.
Registrar staff	As a staff member, I can see a staffing or announcement recommendation together with its evidence and confidence before I act.
Office head	As the office head, I can see weekly volume and feedback trends and a forecast of the next week.

4. Human Task Model

One member role-plays the staff member and handles one typical request aloud (“Can I enroll? I am a transferee and I have my TOR”) while another member records the think-aloud. The steps below are a **draft** that the team must replace and confirm with its own recording.

[Insert 3–5 lines from the team’s recorded think-aloud: date, who role-played, who recorded, and the transcript excerpt.]

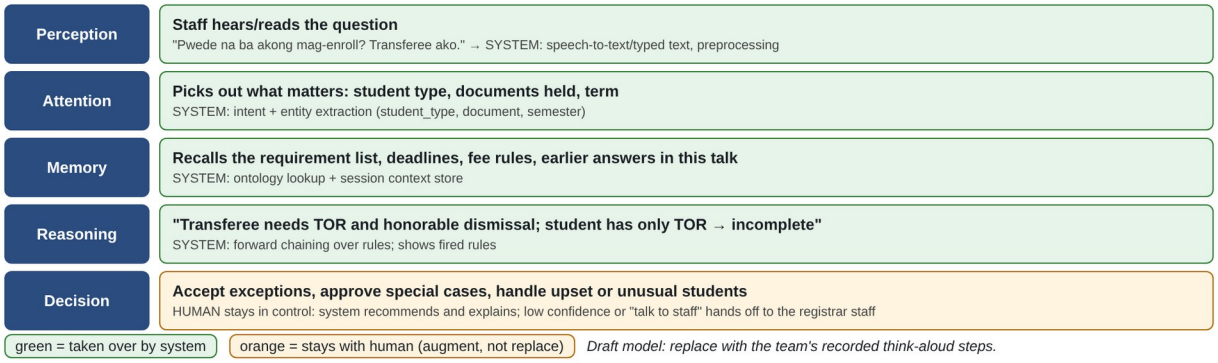


Figure 1. Human task model (perception, attention, memory, reasoning, decision). Green steps are taken over by the system; orange stays with the human.

The system takes over the repetitive recall and checking work (steps 1–4). The human keeps the decision: exceptions, special cases, and upset students stay with the registrar staff. The design therefore augments the staff rather than replacing them.

PART B – THE PROPOSED COGNITIVE SYSTEM

5. System Overview and Why It Must Be Cognitive

Cognitive Campus Assistant is a web application where students ask enrollment and student-service questions in English, Filipino, or mixed language, by typing or speaking. It understands the question, remembers what the student already said, checks the student's situation against a rule base built from the handbook, and answers with a confidence score and the rules and sources behind it. Staff see a dashboard of common concerns with recommendations that show their evidence and confidence.

Characteristics needed.

- **Interactive:** it asks clarifying questions when the question is unclear or a fact is missing.
- **Iterative and stateful:** it remembers the student type and documents already mentioned during the session.
- **Contextual:** answers depend on student type, term, and language (Taglish), not only on keywords.
- **Adaptive:** the intent model is retrained from the failed-query log so that it improves with use.

Why ordinary software is not enough. A fixed menu or keyword search fails on varied phrasing and mixed language, cannot give a confidence score, and cannot learn from missed questions. A conventional program also cannot weigh uncertain evidence for a staff recommendation. At the same time, the enrollment policy is fixed and must be auditable, so the decision part uses explicit rules, and the language part uses a trained model. Combining both is what makes the system cognitive rather than a plain FAQ bot.

Use of a large language model. The core system works with no large language model. The team proposes **no LLM** in the base design; if the instructor approves, one optional use is rewording an answer that the rules and knowledge base already produced, and it will be switchable OFF for the "turn off the LLM" test.

6. Features and Scope

Will build (3 weeks, 4 members)	Will not build
F1 Ask about requirements, schedule, fees, offices, clearance, and prerequisites (10 intents)	Real student records, grades, or login (no personal data)
F2 "Can I enroll?" eligibility check using forward chaining, showing rules fired	Diagnosis, legal, or financial advice; payment processing
F3 Confidence on every answer; clarification and hand-off when low	Native mobile app or Messenger deployment

Will build (3 weeks, 4 members)	Will not build
F4 Voice question input (local Whisper) tested on 20 consented samples	Text-to-speech replies and many-language support (Filipino and English only)
F5 Feedback sentiment classifier (thumbs and short comments)	Automated replies to staff mail or social media
F6 Staff dashboard: 3 descriptive views, a volume forecast, and a recommendation with evidence and confidence	Integration with the school's real information system

7. Architecture

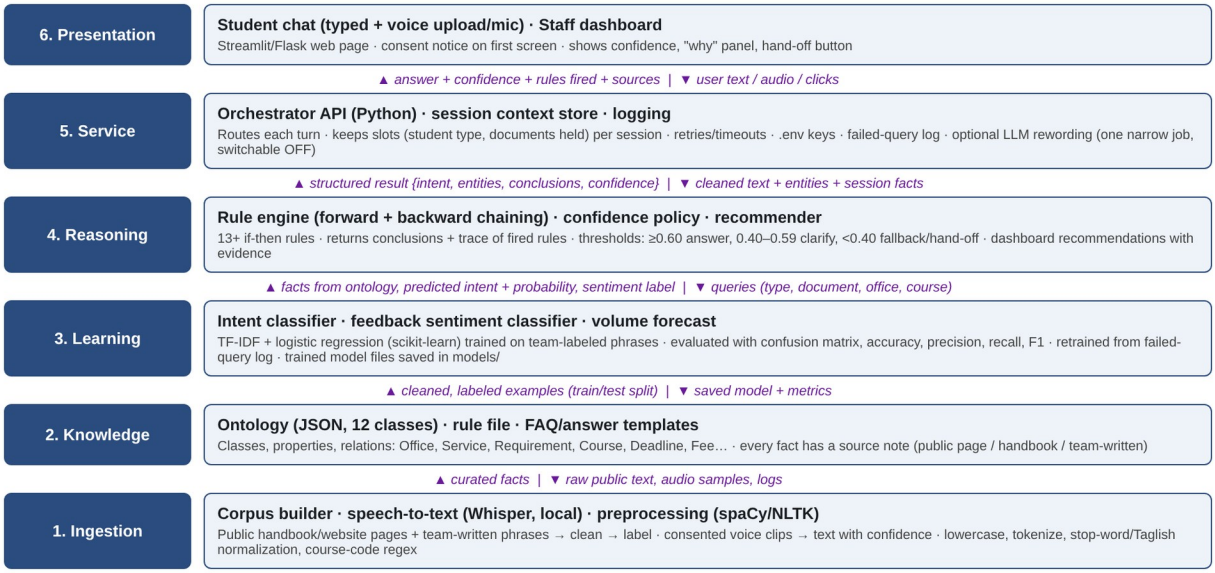


Figure 2. Six-layer architecture with the data crossing each boundary.

Service / library	Layer	Used for	Cost
Python, pandas, scikit-learn	Learning	Intent and sentiment models, forecast	Free
spaCy or NLTK	Ingestion	Preprocessing and entity patterns	Free
Whisper (open source, local)	Ingestion	Speech-to-text; no key needed	Free, runs on laptop
JSON ontology and own rule engine	Knowledge, Reasoning	Facts, rules, chaining trace	Free
Streamlit or Flask	Presentation	Chat page and dashboard	Free
Optional LLM API (only if approved)	Service	Reword an existing answer; key in .env	Free tier; off by default

8. Knowledge Design

Draft ontology (12 classes). Stored as a JSON file in knowledge/ontology.json.

Class	Main properties	Relationships
Student	type, term, documents_held (session only)	hasType → StudentType

Class	Main properties	Relationships
StudentType	name (freshman, transferee, returning)	requires → Requirement
Requirement	name, description, issuing_body	neededFor → Service
Service	name, steps, online_or_window	providedBy → Office; hasFee → Fee; hasDeadline → Deadline
Office	name, location, hours	provides → Service
Course	code, title, units	prerequisiteOf → Course; partOf → Program; offeredIn → Semester
Program	name (e.g., BSCS), max_units	contains → Course
Semester	name, start_date, end_date	has → Deadline
Deadline	label, date	appliesTo → Service
Fee	label, amount, payment_options	chargedFor → Service
Policy	statement, source	appliesTo → StudentType
Announcement	text, date_posted, source	about → Service

All facts are taken from public pages or the handbook, or written by the team, with a source note on each. The requirement lists below are drafts; the team must verify them against the actual handbook of [School name].

First rules (13). Forward chaining over facts such as student_type(transferee) and has(tor). “not” is negation as failure on base facts only (closed-world assumption), which keeps results independent of rule order. This prototype was run and tested (code/rules_trace.py, 6 test cases passed).

#	IF	THEN
R1	student_type(freshman)	needs(form138)
R2	student_type(freshman)	needs(psa_birth_certificate)
R3	student_type(transferee)	needs(tor)
R4	student_type(transferee)	needs(honorable_dismissal)
R5	student_type(returning)	needs(clearance)
R6	needs(D) AND NOT has(D)	missing(D)
R7	missing(D)	docs_incomplete
R8	student_type(freshman) AND has(form138) AND has(psa_birth_certificate)	docs_complete
R9	student_type(transferee) AND has(tor) AND has(honorable_dismissal)	docs_complete
R10	student_type(returning) AND has(clearance)	docs_complete
R11	docs_complete AND fees_paid	eligible(official_enrollment)
R12	docs_complete AND NOT fees_paid	next_step(pay_assessment)
R13	docs_incomplete	next_step(submit_missing_documents)

Forward chaining trace, worked by hand. Facts (made-up student): student_type(transferee), has(tor). In each cycle, all rules are matched against the facts at the start of the cycle.

Cycle	Rule fired	Binding	New fact
1	R3	–	needs(tor)
1	R4	–	needs(honorable_dismissal)
2	R6	D = honorable_dismissal (has(tor) exists, so D = tor does not fire)	missing(honorable_dismissal)
3	R7	D = honorable_dismissal	docs_incomplete
4	R13	–	next_step(submit_missing_documents)
5	none	R8–R12 do not match	stop (fixpoint)

Answer shown: “Submit your honorable dismissal first,” with fired rules R3, R4, R6, R7, R13. **Backward chaining** is used for “why can’t I enroll?”: goal eligible(official_enrollment) needs R11 → docs_complete and fees_paid; docs_complete via R9 needs has(tor) (true) and has(honorable_dismissal) (unknown, so it fails); therefore the blocking fact is the missing honorable dismissal.

9. Data and Corpus Plan

Item	Plan
Sources	Public school website pages, published handbook, and posted announcements (no data requested from any office); phrases and records written by the team; voice clips recorded by consenting teammates and classmates.
Size	Intent corpus: 10 intents × 30 phrases = 300 examples (minimum is 200). Feedback corpus: about 120 short comments labeled positive, neutral, negative. Test set: 50 unseen questions written by a member who did not write the training phrases. Speech: 20 or more clips.
Labels	One intent label per phrase; sentiment label per comment; the reference text of each audio clip for word error rate.
Cleaning	Remove duplicates, lowercase, normalize spelling variants of Taglish (e.g., “po”, “pano”), mask course codes consistently, strip any personal detail; keep a change log.
Split	80% train / 20% test, stratified by label, with a fixed random seed; the 50 unseen questions are never used in training or tuning.
Known gaps	Mostly Filipino and English; few regional-language phrases; speaker accents of the team over-represented. To be extended with classmates’ clips.

A sample of 50 team-written labeled items (5 per intent) is attached in Appendix C and as data/starter_corpus_50.csv. The data card will use the course template and be completed in the Final.

10. Technique Comparison

Accuracy figures are **targets to be measured** in the Week-2 pilot on a small held-out set; no result is claimed yet.

Layer	Options compared	Accuracy	Data needed	Cost	Explainability	Choice and why
Intent recognition	(a) TF-IDF + logistic regression (b) cloud NLU service	(a) to pilot; short in-domain text suits it (b) to pilot	(a) about 30 phrases per intent (b) 10–15 per intent	(a) free (b) free tier with quota	(a) top words per prediction (b) score only	(a): data stays local, probability gives confidence, and every prediction can be explained at the defense
Knowledge storage	(a) JSON ontology (b) OWL in Protégé	Same facts	Same facts	Both free	(a) readable by all members (b) formal reasoning	(a): simpler for four people in 3 weeks; OWL is a stretch goal
Reasoning	(a) forward/backward chaining rules (b) learned classifier for eligibility	(a) exact on policy (tested on 6 cases) (b) depends on data	(a) handbook only (b) labeled eligibility cases	Both free	(a) shows rules fired (b) hard to explain	(a): policy is fixed and must be auditable
Trained model (feedback)	(a) Naive Bayes (b) logistic regression	Both to pilot	About 120 comments	Free	Both show word weights	Chosen after pilot; start with (a) as Bayes-based baseline
Perception	(a) Whisper local (b) cloud speech API	(a) and (b) measured by word error rate on 20 clips	None for training	(a) free (b) free tier	(a) transcript with confidence (b) same	(a): no key or quota and no voice leaves the laptop; accuracy on local names to be tested
Conversation	(a) own dialogue manager in Python (b) Rasa	Both able	Phrases and flows	Both free	(a) transparent (b) more setup	(a): fewer moving parts; Rasa is the fallback
Analytics / recommendation	(a) moving average + rules with evidence (b) ARIMA	(a) to measure by error on held-out weeks	Chat logs (sample)	Free	(a) simple to explain	(a): little data, so a simple, explainable method
Interface	(a) Streamlit (b) Flask + HTML	–	–	Free	–	(a): faster to build both chat and dashboard

11. Conversation Design

Intents (10) and entities (5). Each intent is answered from the rule base, the knowledge base, or a hand-off.

Intent	Example	Handled by
greet	“hi po good morning”	Greeting and privacy notice
ask_requirements	“what do I need to enroll as a freshman?”	Ontology (Requirement)
check_eligibility	“can I enroll? transferee ako, may TOR”	Rules (forward chaining)

Intent	Example	Handled by
ask_schedule_deadline	“when is the last day of enrollment?”	Ontology (Deadline)
ask_fees_payment	“where do I pay my assessment?”	Ontology (Fee)
ask_office_hours_location	“what time does the registrar open?”	Ontology (Office)
ask_course_prereq	“prerequisite of CS 201?”	Ontology (Course)
ask_clearance_process	“how do I get my clearance?”	Ontology (Service)
request_human	“pakausap po sa staff”	Hand-off
give_feedback	“thanks, that was helpful”	Sentiment classifier

Entities: student_type (freshman, transferee, returning), document (TOR, Form 138, clearance, …), office, course_code (pattern such as CS 201), semester or date. **Preprocessing:** lowercase, tokenize, normalize Taglish particles, extract entities with patterns and a dictionary.

Confidence policy: probability ≥ 0.60 answers; 0.40–0.59 asks a clarifying question with buttons; below 0.40 triggers the fallback message and, after two misses, a hand-off. Every answer shows its confidence.

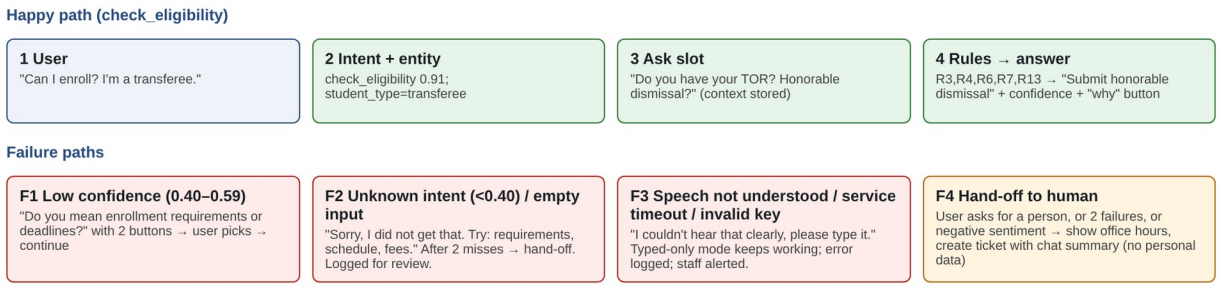


Figure 3. Conversation flow: happy path and failure paths.

Script (happy path). Student: “Can I enroll? Transferee ako.” → System (check_eligibility 0.91, student_type=transferee): “Do you have your TOR? Your honorable dismissal?” → Student: “TOR lang.” → System: “Not yet ready. Submit your honorable dismissal first. Why this answer? Rules R3, R4, R6, R7, R13.” → Student: thumbs up (feedback logged).

Failure paths. (F1) Low confidence: “Do you mean requirements or deadlines?” (F2) Unknown or empty input: a helpful prompt, logged for review, and a hand-off after two misses. (F3) Speech not understood, API timeout, or invalid key: the system says so, falls back to typed input, and logs the error. (F4) Hand-off: the student asks for a person or sentiment turns negative, and the system shows office hours and creates a ticket with a chat summary without personal data. Context (student type, documents held) is remembered within the session.

12. Screen Mockups

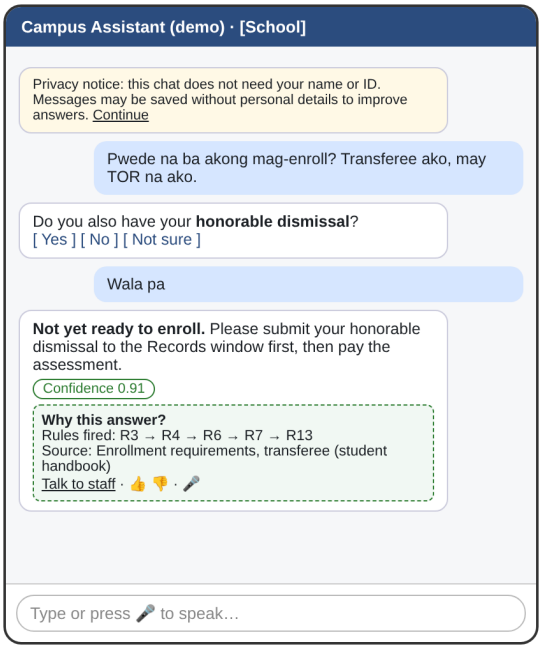


Figure 4. Student chat mockup with a recommendation shown with its confidence, rules, and source.

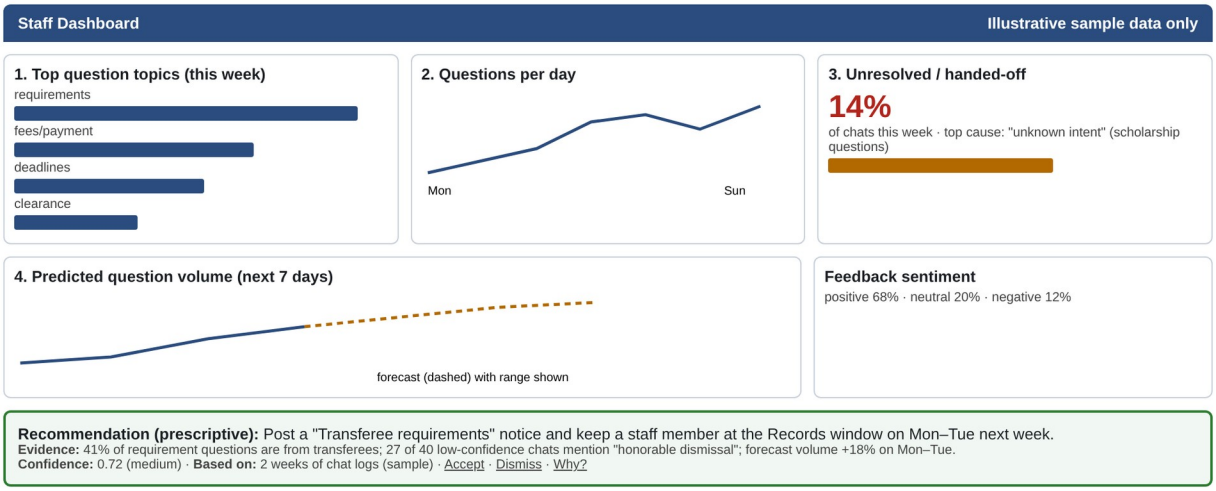


Figure 5. Staff dashboard mockup (illustrative sample data): three descriptive views, a forecast, and a recommendation with evidence and confidence.

13. Evaluation Plan

Component	Metric	Target	Test data
Intent model	Macro F1 (plus confusion matrix, accuracy, precision, recall)	≥ 0.80 on 50 unseen questions; retrained at least once with results documented	50 unseen questions
Confidence	Accuracy of answers at confidence ≥ 0.60	≥ 85%	Held-out set

Component	Metric	Target	Test data
Rules	Correct conclusion and fired-rule trace	100% of 20 hand-made cases	20 cases written by the team
Sentiment	Accuracy and per-class F1	≥ 0.75 accuracy	20% held out
Speech	Word error rate	< 20% on 20 or more local clips	Consented clips
Conversation	Task completion and failure paths handled	≥ 3 failure paths work; hand-off works in 100% of tries	Peer test script
Failure handling	Empty input, unknown intent, API timeout, invalid key	All 4 handled gracefully and logged	Failure drill
Dashboard forecast	Mean absolute percentage error	≤ 25% on held-out weeks	Sample logs
Fairness (Final)	Accuracy gap between 2 groups (e.g., English vs. Taglish; clear vs. noisy audio)	Gap ≤ 10 points, or mitigation applied	Test set split by group

14. Preliminary Privacy and Risk Plan

- **Personal data touched:** typed questions (may contain a name by accident), voice clips of teammates, and chat logs. The system never asks for a name, ID number, grades, or contact details, and all records used in development are made up.
- **Consent:** a notice on the first screen states purpose and storage; a signed consent form is collected from every person whose voice is recorded; no health, religious, or financial details are collected.
- **RA 10173 principles:** transparency (notice and "why this answer"), legitimate purpose (answering enrollment questions only), proportionality (collect only what is needed; mask or remove personal details from logs; keep raw logs for a short, stated period of [days, agreed with instructor]).
- **Who could be treated unfairly:** students who write in Taglish or regional languages, speakers with accents or noisy audio, and first-generation students unfamiliar with office terms. Mitigations planned: more varied training phrases, test accuracy by group, human hand-off always available.
- **Other risks:** wrong policy information (mitigated by source notes and instructions to confirm with staff), API keys leaking (kept in .env and never committed), and the LLM answering everything (not allowed; off by default).

15. Work Split

Member	Final build responsibility
A [Name] – Knowledge and conversation	Ontology, rules, inference engine, intents, entities, dialogue flows, conversation design chapter

Member	Final build responsibility
B [Name] – Data and learning	Corpus, cleaning, labeling, model training and retraining, bias check, data card, evaluation report
C [Name] – Integration and interface	Orchestrator, keys, errors, logging, chat interface, installation guide
D [Name] – Perception and analytics	Speech input and 20-sample test, dashboard, recommender with evidence and confidence, user manual
All members	Architecture, privacy safeguards, peer testing, pull-request reviews, presentation

Every member must be able to explain and change every part of the system on the spot.

16. APPENDICES

Appendix A – AI Use Declaration

Team: [Team name]

Tool	Used for	Part of the project	Member responsible
Claude (Anthropic)	Drafted the proposal text and tables; generated the figure mockups; wrote and ran the draft rule-engine prototype and the starter phrases	Whole proposal; code/rules_trace.py; data/starter_corpus_50.csv	[Name]
[Other tool, if any]	[purpose]	[section]	[Name]

We declare that we reviewed, understood, and can explain and change every part of this submission, including any part produced with AI assistance.

Signed: [all four members]

Appendix B – Consultation Minutes

Consultation	Date and attendees	Topics discussed	Decisions and action items
Prefinal 1	[date; members present]	[topics]	[decisions]
Prefinal 2	[date; members present]	[topics]	[decisions]

Appendix C – Corpus Sample (50 items, team-written drafts)

ID	Text	Intent label
S001	hello	greet
S002	hi po good morning	greet
S003	good afternoon po	greet
S004	hey, may I ask something?	greet
S005	kumusta po	greet
S006	What documents do I need to enroll as a freshman?	ask_requirements
S007	ano po ang requirements for transferee?	ask_requirements
S008	What do I need to bring to the registrar for enrollment?	ask_requirements
S009	Requirements for returning students po?	ask_requirements
S010	Do I need a PSA birth certificate to enroll?	ask_requirements
S011	Can I enroll now? I am a transferee and I have my TOR	check_eligibility
S012	I'm a freshman, I already have Form 138 and PSA, am I ready to enroll?	check_eligibility
S013	Am I eligible to enroll as a returning student?	check_eligibility
S014	Check if my documents are complete	check_eligibility
S015	pwede na ba ako mag-enroll, wala pa akong clearance	check_eligibility
S016	When is the last day of enrollment?	ask_schedule_deadline
S017	What is the schedule for late enrollment?	ask_schedule_deadline

ID	Text	Intent label
S018	Kailan ang start ng classes this semester?	ask_schedule_deadline
S019	When is the deadline for adding and dropping subjects?	ask_schedule_deadline
S020	What date is the payment of fees?	ask_schedule_deadline
S021	How much is the tuition for a BSCS student?	ask_fees_payment
S022	Where do I pay my assessment?	ask_fees_payment
S023	Pwede po ba installment ang payment?	ask_fees_payment
S024	What are the payment options at the cashier?	ask_fees_payment
S025	Magkano ang miscellaneous fee?	ask_fees_payment
S026	What time does the registrar open?	ask_office_hours_location
S027	Where is the library located?	ask_office_hours_location
S028	Anong oras po ang office hours ng guidance office?	ask_office_hours_location
S029	Saan ang registrar's office?	ask_office_hours_location
S030	Is the registrar open on Saturday?	ask_office_hours_location
S031	What is the prerequisite of CS 201?	ask_course_prereq
S032	Can I take Data Structures if I haven't passed Programming 1?	ask_course_prereq
S033	anong prerequisite ng CS ELEC 6?	ask_course_prereq
S034	Which subjects should I take in 2nd year first semester?	ask_course_prereq
S035	How many units can I take this semester?	ask_course_prereq
S036	How do I get my clearance?	ask_clearance_process
S037	What are the steps to get a clearance for graduation?	ask_clearance_process
S038	Ano po ang process ng clearance sa library?	ask_clearance_process
S039	Who signs the clearance form?	ask_clearance_process
S040	Do I need clearance before enrolling again?	ask_clearance_process
S041	I want to talk to a staff member	request_human
S042	Pakausap po sa registrar staff	request_human
S043	Can you connect me to a real person?	request_human
S044	human agent please	request_human
S045	I need help from the office directly	request_human
S046	Thanks, that was very helpful!	give_feedback
S047	The answer was wrong, I wasted my time	give_feedback
S048	Ang bilis ng sagot, salamat po	give_feedback
S049	This is confusing and not useful	give_feedback
S050	Great assistant, easy to use	give_feedback

Source for all items: written by the team for this project (no real people, no data from any office). Status: to be reviewed and extended to 300 items by Member B.

Appendix D – Candidate Situations Submitted

Rank	Situation	Why
1	Cognitive Campus Assistant (enrollment and student services)	Public information is easy to find; we are clients of the service
2	[Candidate 2]	[reason]
3	[Candidate 3]	[reason]